

From Generative to Embodied Intelligence: Network Evolution for Large-AI-Model Agents

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Abstract—Large AI models (LAMs) are evolving from content generators to tool-using agents and embodied systems that interact with the physical world. This evolution changes both network traffic and the consequences of delayed or incorrect information. This survey develops a network-centric comparison of generative, agentic, and embodied LAMs, linking model metrics and benchmarks to communication requirements. Generative services create asymmetric, bursty multimodal demand; agentic workflows introduce repeated context exchange, branching, and retry-amplified load; embodied systems couple sensing and action traffic through closed-loop deadlines and control performance. We compare scenario-dependent quality-of-service requirements, trace architecture evolution from cloud-centric inference to stateful edge orchestration and hierarchical control, and organize enabling technologies across token, transport, and coordination planes. A prototype case study evaluates blind recognition of 2,300 individual 10-s traffic segments using only packet timing, size, and direction. On the 460-segment test set, statistical linear, convolutional, and bidirectional recurrent classifiers achieve 99.83–100% Macro-F1, demonstrating distinguishable traffic dynamics under the evaluated capture conditions. A complementary, proposed active recognition stage uses endpoint-declared descriptors for QoS mapping. Finally, we identify tokenizer compatibility, token-native protocols, and joint network–model–control evaluation as open directions for connecting distributed intelligence.

Index Terms—Agentic AI, embodied AI, edge intelligence, large AI models, multi-agent systems, networked control, semantic communications, token communications, ultra-reliable low-latency communications.

I. INTRODUCTION

LARGE AI models (LAMs) are extending networked services from content generation to autonomous task execution and physical interaction. Generative systems produce text, images, audio, and video [1]; agentic systems invoke tools, retrieve memory, and coordinate multiple models; embodied systems couple multimodal perception with physical actuation. Here, LAM denotes foundation-model-based systems, including large language models (LLMs), multimodal large language models (MLLMs), world models, and vision–language–action (VLA) models. The stages describe overlapping capabilities, not disjoint model families.

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Existing research provides foundations for this evolution. Autoregressive language and text–image models enable scalable token-based generation [2], [3], while MMLU and BIG-bench evaluate capabilities across diverse tasks [4], [5]. Agent-Bench extends evaluation to tool-using trajectories [6], and ReAct and Reflexion incorporate reasoning, action, feedback, and retry [7], [8]. Meta-World, RT-2, and LIBERO further bring manipulation and sensorimotor interaction into the evaluation of embodied systems [9]–[11]. The unit of success thus changes from generated content to a completed task and then to a physical episode, motivating a corresponding comparison of network requirements.

Network measurements already indicate the importance of looking beyond downlink-dominated content delivery. Ericsson’s June 2026 Mobility Report finds that uplink traffic grew faster than downlink traffic at 43 of 55 measured service providers during 2025 [12]. This is an aggregate network observation, not a measurement of LAM traffic alone. Separately, its urban-area scenario analysis projects 2031 uplink traffic at three and five times the 2025 level under medium and high AI-adoption assumptions, respectively [12]. These conditional projections, rather than guaranteed growth rates, motivate examining how multimodal uploads, agent interactions, and sensor streams change network load and its temporal structure.

Standardization is recognizing these changes. ITU-R Recommendation M.2160-0 identifies artificial intelligence and communication as one of six IMT-2030 usage scenarios, encompassing distributed inference, computation offloading, and computing-resource orchestration [13]. The 3GPP study on 6G use cases and service requirements includes AI-agent communication and network-supported AI/ML training and inference [14]. Together, these documents establish a direction for supporting distributed intelligence, while leaving the service-specific mapping from model behavior to end-to-end network requirements to be developed; they do not prescribe a universal latency budget for all LAMs.

Collaborative research projects are translating this direction into system-design agendas. Hexa-X-II (2023–2025) targeted a 6G platform blueprint combining connectivity with sensing, computing, and trustworthy AI, together with system- and component-level validation [15]. The 6G-GOALS project (2024–2026) investigates semantic and goal-oriented communication, including distributed reasoning and time-sensitive information exchange [16]. Their complementary emphases on integrated infrastructure and task-relevant transmission motivate a joint view of what intelligence needs to communicate and how networks should support it.

Industry initiatives further connect these research directions

to deployment. The AI-RAN Alliance, which reported 132 members in February 2026, distinguishes AI-based RAN optimization, shared AI/RAN computing, and edge-hosted AI applications through its AI-for-RAN, AI-and-RAN, and AI-on-RAN activities [17]. The latter two are particularly relevant to networks that host and carry LAM workloads, rather than only use AI to optimize radio operation. In parallel, the Linux Foundation’s Agent2Agent (A2A) project promotes interoperability among agents across platforms and vendors [18], while NVIDIA’s open GR00T N1 model targets generalist humanoid capabilities [19]. These efforts motivate considering distributed inference, cross-agent exchanges, and physical interaction within a common service-oriented network perspective.

Distributing these capabilities across devices, edge servers, and clouds creates distinct resource burdens. Generative services combine modality-dependent uplink/downlink asymmetry with bursty transfers whose volume depends on content fidelity. Agentic services repeatedly exchange context and tool results, so branching and retries increase traffic and task-completion time. Embodied services share a closed-loop deadline among sensing, communication, inference, and action; additional sensor detail can improve decisions while consuming the time available to execute them. These demands require network design to consider directional load, task-level reliability, and timely physical feedback alongside communication and computing costs, rather than average throughput alone.

Several surveys provide complementary foundations. Xi *et al.* review LLM-based agent construction, applications, and agent societies [20]; Y. Liu *et al.* organize embodied AI around robots and simulators, perception, interaction, agents, and sim-to-real adaptation [21]. These model- and application-oriented perspectives explain how intelligence is constructed and evaluated. On the networking side, Qu *et al.* survey edge LLM caching, delivery, training, and inference [22]. Guo *et al.* examine semantic communication networks for multi-agent interaction, including architecture, security, and privacy [23]; Z. Liu *et al.* survey integrated sensing and edge AI, including joint sensing–communication–computation design and its metrics [24]. Semantic and token communication frameworks further connect transmitted representations to information relevance [25], [26].

These works already address distributed intelligence and task-oriented communication, but organize the literature around different system boundaries: agent construction, model deployment, semantic networking, or intelligent perception. Our complementary focus is the change in the communication problem across LAM capabilities: from delivering generated content, through completing a branching tool-using workflow, to maintaining timely sensing and action. Efficient delivery of one generative response does not by itself ensure workflow completion or successful physical control. Comparing traffic structure, failure consequences, and evaluation criteria across these stages helps identify which existing mechanisms can be reused and which require adaptation.

Motivated by this transition, we connect model objectives and benchmarks to traffic, quality of service (QoS), archi-

tectures, and enabling technologies within one generative–agentic–embodied framework. Numerical QoS scales are interpreted as deployment-dependent scenarios. This perspective supports both assessment of existing techniques and identification of research gaps. The central design question is: *which information must be delivered, to which model or actuator, by what deadline, and with what task-level loss when resources are limited?*

A. Contributions

The survey makes four contributions.

- It synthesizes generative, agentic, and embodied systems through a network-centric comparative framework that jointly classifies model objectives, token sources, interaction patterns, benchmarks, failure consequences, and network operating bottlenecks.
- It maps AI-side evaluation to network-side quality of service (QoS). The comparison connects content fidelity and bursty directional demand, task success and retry-amplified tail latency, and physical performance under joint loop-delay, jitter, and reliability constraints. Numerical QoS anchors are interpreted as illustrative deployment scenarios.
- It organizes enabling technologies into token, transport, and coordination planes and maps them back to the QoS and topology of each stage. For embodied agents specifically, the synthesis exposes two transfer gaps: compression is commonly perception-oriented rather than control-oriented, and deterministic transmission is commonly source-independent rather than task-coupled.
- It presents a two-stage traffic-recognition case study. The measured passive stage evaluates statistical linear, convolutional, and bidirectional recurrent classifiers on 2,300 individual 10-s segments, obtaining 99.83–100% test Macro-F1 without payload or application identifiers. The proposed active stage complements this coarse traffic classification with authenticated endpoint descriptors for stage-specific QoS mapping; it is not part of the reported classification experiment.

Section II reviews model evolution and benchmarks. Sections III and IV study metric and architecture evolution, respectively. Section V surveys enabling technologies, Section VI presents the case study, and Section VII discusses open challenges. Figure 1 summarizes this organization.

II. FROM GENERATIVE TO EMBODIED LAMS: PERFORMANCE METRICS, BENCHMARKS, AND MODEL EVOLUTION

A. Generative LAMs

Generative LAMs learn a conditional distribution over token sequences [2], [3]. Text is represented by subword tokens; images and video by spatial or spatiotemporal patches or discrete latent codes; and audio by waveform, spectral, or codec tokens. Transformer attention supplies a shared processing interface, but it does not make these tokens computationally equivalent. A text token may represent a word fragment, whereas a visual

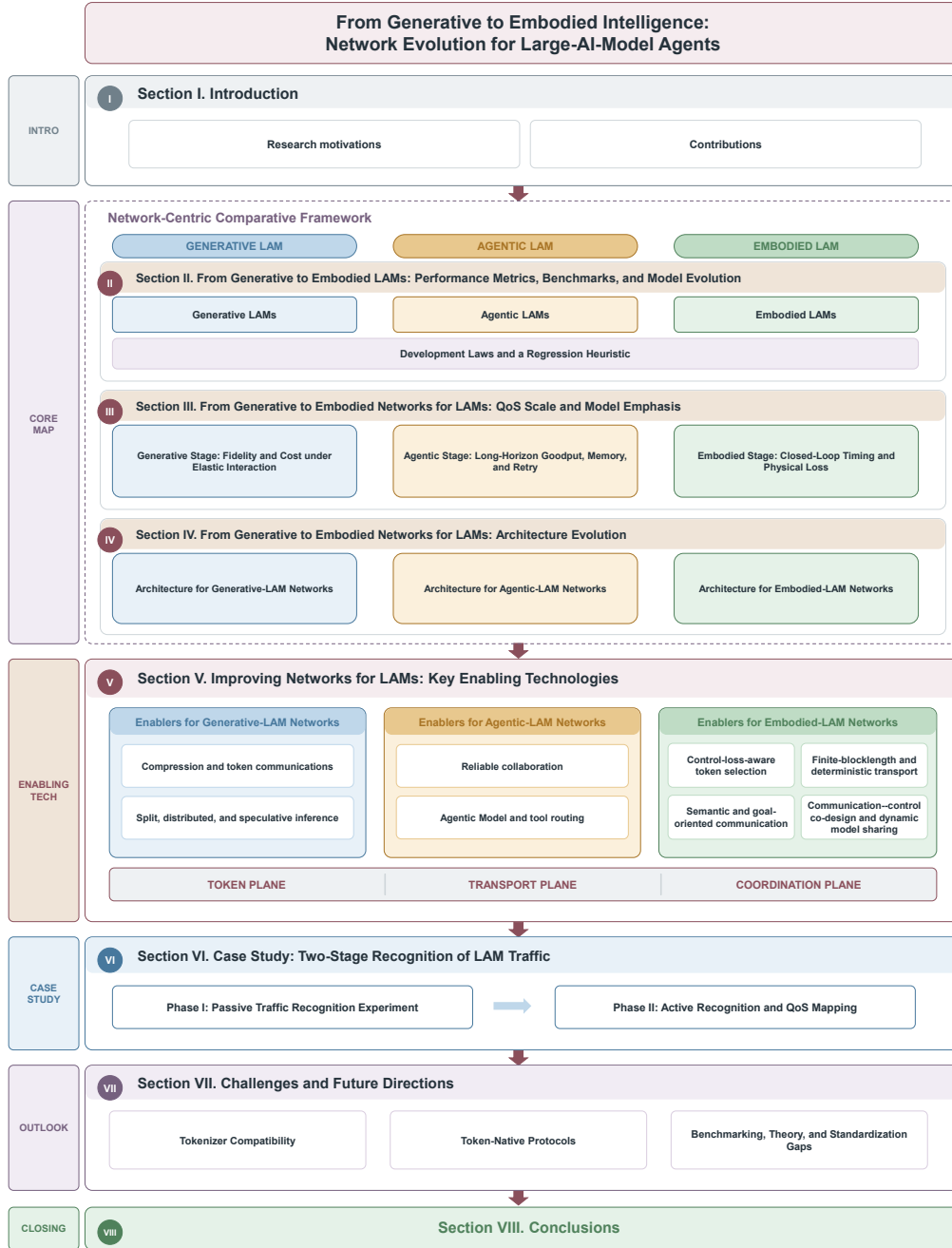


Fig. 1. The outline of this survey.

token represents a local patch and an audio token represents a time interval.

Model-side evaluation in this stage is largely *content-centric*. Perplexity measures predictive fit [2]; exact-match or multiple-choice accuracy measures knowledge and reasoning [4]; and BLEU and ROUGE compare generated and reference text [27], [28]. MMLU spans 57 subjects and tests broad academic knowledge [4]; BIG-bench uses more than 200 heterogeneous tasks to probe capabilities and scaling behavior [5]. For open-ended and multimodal generation, human preference, learned reward, semantic similarity, factuality, and safety are required because literal word or phrase matching cannot

capture meaning, visual quality, or factual correctness.

The token workload depends strongly on both modality and direction. Text understanding performs a parallel prefill over the prompt and retrieved context, whereas text generation is autoregressive: it adds one token per decoding step and grows the key-value cache with sequence length. Image understanding processes a spatial patch grid [29], while diffusion-based image generation repeatedly evaluates a denoising model [30]. Video adds a temporal token dimension to image processing, so its cost scales with frame count as well as resolution; audio understanding and generation operate on dense time-indexed spectral or codec tokens and often require causal streaming.

Consequently, long text output is dominated by sequential decoding, high-resolution image synthesis by repeated denoising, and video generation by the largest combined compute and memory footprint. These modality-dependent workloads define the relevant local model characteristics in this section.

B. Agentic LAMs

An agentic LAM places a large AI model inside a feedback loop with a planner, tool interfaces, memory, verification, and sometimes other agents [7], [8], [31]. The unit of work is no longer one output sequence but a trajectory

$$\tau = (o_0, a_0, o_1, a_1, \dots, o_K),$$

where observations o_k may be web pages, application screenshots, database results, code-execution logs, or messages from peer agents, and a_k may call an API or change external state. Reasoning techniques such as chain-of-thought [32] and tree/graph search [33], retrieval-augmented generation [34], model context protocols [35], and multi-agent debate [31] increase capability by introducing more rounds and more state.

Consequently, evaluation shifts from answer similarity to *process-grounded task completion*. AgentBench, for example, evaluates LLM agents across operating-system, database, knowledge-graph, gaming, household, web-shopping, and web-browsing environments [6]. WebArena tests goal completion in realistic websites [36], whereas SWE-bench evaluates repository-level issue resolution with executable tests [37]. Across these benchmarks, relevant metrics include task success rate, number of tool calls, constraint violations, plan length, recovery from tool failure, long-horizon consistency, and cost per completed task. An agent that produces fluent intermediate text but executes the wrong transaction has failed.

Trajectory reliability compounds over the sequence. If step k succeeds with conditional probability p_k , a simplified serial workflow has

$$P_{\text{task}} = \prod_{k=1}^K p_k, \quad P_{\text{fail}} \leq \sum_{k=1}^K (1 - p_k), \quad (1)$$

where the equality assumes that each p_k already conditions on prior success. Even modest per-call error becomes material for large K . Reason-act traces and verbal feedback enable exception handling and retry [7], [8], but repeated attempts expand context, consume tokens, and create a heavy tail in job completion time.

C. Embodied LAMs

Embodied LAMs ground language and vision in state estimation, world prediction, planning, and action [10], [38], [39]. Architectures include VLA models, latent world models, bird's-eye-view transformers, and joint-embedding predictive architectures. The observation o_t can combine RGB/depth images, point clouds, force/torque, inertial measurements, joint encoders, and messages from other robots. The output u_t can contain waypoints, discrete skills, joint targets, or continuous control actions. Generalist humanoid models such as GR00T N1 illustrate a trend toward shared representations across embodiments and tasks [19].

The benchmark moves into a physical or high-fidelity simulated loop. Meta-World contains 50 manipulation tasks and evaluates multi-task generalization [9]; RT-2 tests seen and unseen skills and semantic generalization on real robots [10]; and LIBERO measures knowledge transfer across lifelong manipulation suites [11]. These widely used benchmarks primarily score success or return under a benchmark-defined horizon and do not standardize end-to-end inference latency. HumanoidBench includes locomotion, stabilization, and whole-body manipulation and runs its 61-dimensional policy at 50 Hz [40], while Habitat 3.0 adds dynamic human collaborators and human-in-the-loop evaluation [41]. ReflexBench directly exposes reaction time: its six dynamic manipulation tasks sweep configurable latency under synchronous and asynchronous inference [42]. Accordingly, balance-sensitive, non-stationary, and multi-actor evaluation should report task outcome jointly with observation-to-action latency and effective control frequency, rather than treating a fixed simulator step as evidence of real-time operation. World models used for control are ultimately judged by episodic return and planning quality, not merely reconstruction [39]; a small perception error may be harmless in one state but decisive near a contact transition, so average token distortion alone is inadequate.

Embodied work uses task-specific objectives and reports task success, episodic return, and trajectory or control error [10], [39], [40]. A benchmark-specific finite-horizon control cost can be written as

$$J_c = \mathbb{E} \left[\sum_{t=0}^H \gamma^t (\|x_t - x_t^*\|_{\mathbf{Q}}^2 + \|u_t\|_{\mathbf{R}}^2) \right], \quad (2)$$

where x_t is the physical state, x_t^* the task-dependent target, u_t the action, and $\mathbf{Q}$ and $\mathbf{R}$ weight state-tracking and control effort. This is not proposed as a universal new score: each benchmark must publish its horizon, weights, termination conditions, and component metrics separately.

Embodied LAM workloads contain continuous, coupled perception and action streams. The model must understand scene semantics, object relations, and task intent from rich multimodal observations while updating actions in real time; adding more frames or viewpoints improves grounding but increases token count and inference cost. This creates a three-way local resource tension. Increasing resolution, context length, or temporal history can improve scene and state estimation, but consumes more operations and memory and lengthens observation-to-action inference. Reducing precision, visual tokens, or action updates saves compute and shortens latency, but can lower task accuracy or make the controller miss a short transition. Increasing the update rate tightens the per-cycle compute budget while repeating multimodal fusion more often. These constraints are most severe in balance, contact-rich, and dynamic multi-actor tasks.

D. Development Laws and a Regression Heuristic

Table I reveals four recurring development laws.

- Interaction frequency and coupling increase: a prompt is weakly coupled to the next request, an agent trajectory is logically coupled across steps, and an embodied trajectory

TABLE I
MODEL AND WORKLOAD EVOLUTION ACROSS LAM STAGES

Stage	Primary objective	Token/source structure	Representative evaluation	Interaction/workload pattern	Failure consequence
Generative	Produce useful and faithful digital content	Subwords, image/video patches, audio or codec tokens; statistical and cross-modal redundancy	Perplexity, accuracy, MMLU, BIG-bench, BLEU/ROUGE, preference, factuality	Prompt/context prefill followed by autoregressive decoding; modality-dependent token count; one/few-shot interaction	Hallucinated or degraded content; often detectable, retryable, and non-physical
Agentic	Complete a goal through reasoning, tools, memory, and collaboration	Generated tokens plus UI frames, logs, API objects, retrieved memory, and inter-agent messages; workflow-induced redundancy	Task success, tool correctness, constraint violations, calls/steps, recovery, cost per completed job	Stateful multi-round graph with fan-out, retries, persistent context, and long-tail completion time	Wrong external operation or cascading plan failure; compensation may be expensive or impossible
Embodied	Perceive, predict, and act effectively in the physical world	Multiview/multimodal perception, world-state and action tokens; spatial, temporal, task, and physical coupling	Task success, episodic return, action/trajectory error, world-model error, planning quality, sim-to-real gap, control cost, inference latency	Continuous, bidirectional, multisource loops at heterogeneous periods; scene understanding and action updates at heterogeneous frequencies	Task failure, unstable motion, or divergence from the commanded state; late information can be worse than missing information

is both logically and dynamically coupled through the environment.

- Failure propagates farther: a generative error usually affects one response, an agentic error can corrupt the remaining task graph and external state, and an embodied error changes the next observation through physical dynamics.
- The computational path evolves from one-pass generation, to iterative stateful trajectories, and then to continuous multi-rate perception–action loops. The dominant bottleneck consequently shifts from sequential decoding and key–value memory, to repeated inference and context expansion, and finally to concurrent multimodal fusion under a per-cycle deadline.
- The exploitable structure becomes richer: generative tokens mainly expose statistical redundancy, agentic trajectories add repeated context and workflow redundancy, and embodied streams add geometry-, kinematics-, and task-induced constraints. Selective computation and reuse therefore become more structured, but their validity must be judged at the response, trajectory, or closed-loop episode level.

An intuitive, but deliberately non-normative, interpretation is a *regression heuristic*. Artificial systems have scaled capabilities in an order that is roughly opposite to the layering of human competence: codified knowledge and content generation first, explicit planning and tool use next, and robust sensorimotor interaction last. Conversely, loss of higher explicit reasoning can leave basic procedural or reflexive behavior intact. The analogy helps explain why generating fluent text does not imply reliable tool use, and why either capability does not imply safe physical intelligence. It is neither a biological law nor an originality claim; its engineering implication is only that later stages inherit earlier representation and generation abilities while adding state, timing, interaction, and physical grounding. And the embodied system requires more sophisticated training and ingenious system optimization.

III. FROM GENERATIVE TO EMBODIED NETWORKS FOR LAMS: QOS SCALE AND MODEL EMPHASIS

The three stages impose different quantitative requirements on common network indicators, including latency, bit and token rates, traffic and connection densities, reliability, etc. Generative services combine elastic completion times with highly variable multimodal rates; agentic services tighten per-step reliability and accumulate long-tail task delay; embodied services require synchronized bidirectional delivery under strict loop deadlines and dense sensing loads. Table II¹ gives a scenario envelope that instantiates the radar-style comparison motivating this survey.

A. Generative Stage: Fidelity and Cost under Elastic Interaction

The content-centric benchmarks in Section II first determine whether an output is useful. Delivering that quality produces modality-dependent bit and token rates: Table II illustrates 10^3 token/s and a peak 0.5-Gbit/s multimodal flow. Text requests typically produce small uplink prompts followed by longer downlink responses, whereas image understanding requires substantial input uploads and media generation produces large output downloads. The uplink/downlink ratio therefore varies across tasks; it cannot be represented by one fixed traffic profile. Higher output fidelity and longer responses further increase payload size and transient bandwidth demand.

This signature burdens a current mobile network through variability and burstiness. User arrivals can become correlated after a shared event, whereas prompt length, output modality, and final response length are not fully known at admission.

¹Two cautions are essential. First, multiplying every maximum in one row by every maximum in another creates an impossible workload; duty factors and statistical multiplexing must be declared. Second, a “token” is not a fixed number of bits across text, images, video, latent states, or actions. Token rate therefore complements rather than replaces bit rate. The assumptions in Table II make sensitivity analysis possible: changing r_0 , N_{par} , L_{state} , a , or trajectory length H immediately produces a new deployment-specific envelope.

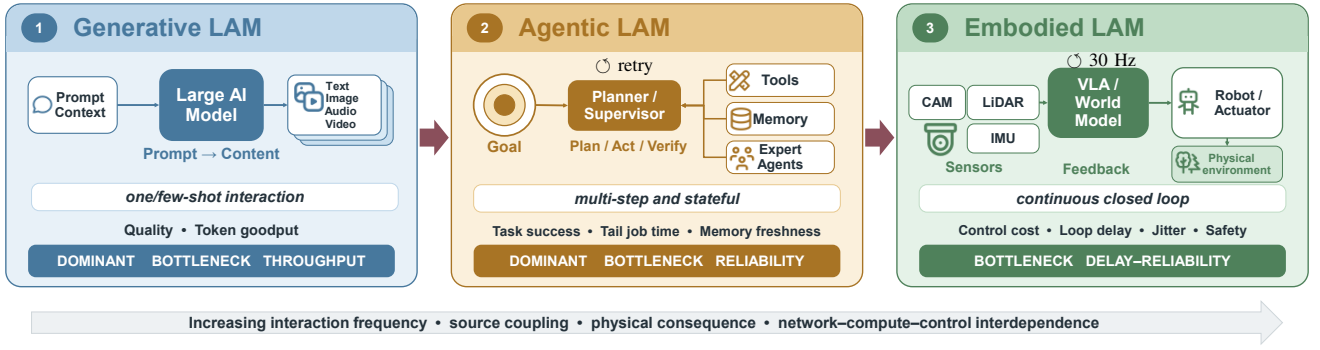


Fig. 2. Evolution of LAM services and their dominant network operating bottlenecks. Requirements accumulate rather than replace one another: embodied systems still need throughput and semantic reliability, but must add deterministic closed-loop behavior. The circular marker on the planner/supervisor denotes retry; each embodied sensing-action loop carries a task-specific update rate, with 30 Hz shown as an illustrative requirement.

TABLE II
ORDER-OF-MAGNITUDE QOS ANCHORS AND EXPLICIT SCENARIO BASIS

Common network quantity	Generative	Agentic	Embodied	Interpretation/calculation basis
Network contribution to latency	100 ms	10 ms	1 ms	Illustrative budget after separating model execution: interactive response, multi-step coordination, and fast physical loop, respectively. Full end-to-end latency still follows (3).
Peak service-token rate	10^3 token/s	5×10^4 token/s	10^5 token/s	$R_{\text{tok}} = N_{\text{par}} r_0$, with source-equivalent stream rate $r_0 = 10^3$ token/s and parallel context/agent/modality counts $N_{\text{par}} = 1, 50, 100$.
Peak service bit rate	0.5 Gbit/s	5 Gbit/s	10 Gbit/s	A high-fidelity multimodal stream is the 0.5-Gbit/s unit; agentic fan-out and embodied multiview sensing aggregate roughly 10 and 20 such units. Compression quality remains model- and modality-dependent [22], [43].
Active-token density	10^7 token/km ²	10^8 token/km ²	10^9 token/km ²	$D_{\text{tok}} = N_c L_{\text{state}}$ using $L_{\text{state}} = 10^3$ active tokens per endpoint and the connection-density anchors below.
Area traffic density	10 Tbit/s/km ²	100 Tbit/s/km ²	1000 Tbit/s/km ²	$D_b = 10^{-3} N_c m a r_b$, where r_b is in Gbit/s. The tuples (m, a, r_b) are (2, 1, 0.5), (1, 0.2, 5), and (1, 0.1, 10); m and a denote parallel flows and active duty factor.
Step/deadline reliability	99%	99.9%	99.999%	$p_{\text{step}} \geq P_{\text{task}}^{1/H}$: $(P_{\text{task}}, H) = (0.99, 1)$, $(0.99, 10)$, and $(0.999, 100)$ yield approximately the listed anchors.
Connection density	10^4 /km ²	10^5 /km ²	10^6 /km ²	Tenfold scenario scaling from personal generative sessions, through dense agent/tool endpoints, to massive sensors and physical agents; the top anchor matches the established massive-connectivity order [14].

Small text exchanges coexist with large media uploads or downloads, causing rapid changes in directional load. Fixed uplink/downlink allocation can leave one direction underused while the other is congested; peak-rate reservation wastes capacity between bursts, while average-rate provisioning cannot absorb coincident media transfers. The relevant network burden is thus asymmetric, bursty resource demand, characterized by directional traffic ratios, peak-to-mean load, burst duration, and useful-token goodput.

B. Agentic Stage: Long-Horizon Goodput, Memory, and Retry

The process-grounded benchmarks in Section II make the complete K -step task, not an individual model response, the unit of success. Their task-success, tool-correctness, attempts, recovery, and cost metrics translate into end-to-end job time, per-step reliability, context freshness, active state, and useful task-level goodput. For example, a 99% target over ten serial steps requires $p_{\text{step}} \geq 0.99^{1/10} = 99.8995\%$, which motivates the 99.9% illustrative anchor in Table II. Its 10-ms network budget, 5×10^4 token/s peak rate, and 10^5 /km² connection density represent multiple concurrent agent, tool, memory, and verifier exchanges, not a uniform rate sustained by every task. Longer plans and more branches tighten the reliability

requirement and broaden both state size and completion-time distributions.

This numerical signature produces a different load on a mobile network. Task graphs alternate silent tool execution with synchronized fan-out and fan-in; timeouts, debate, and retries create microbursts, duplicate state, and a heavy tail of unfinished jobs. The set of active flows changes during execution, so admission and scheduling cannot rely on a fixed session graph or known payload. Moreover, packet success is insufficient when an observation is stale, a tool operation is duplicated, or a schema is inconsistent. The network must therefore measure tail task-completion time, state freshness, retry-amplified load, and completed-task goodput, while isolating correlated bursts from other latency-sensitive services.

C. Embodied Stage: Closed-Loop Timing and Physical Loss

Embodied intelligence inherits generative fidelity and agentic trajectory reliability, then adds sensing, actuation, and a physical deadline:

$$T_t^{\text{loop}} = T_t^{\text{sense}} + T_t^{\text{enc}} + T_t^{\text{ul}} + T_t^{\text{queue}} + T_t^{\text{inf}} + T_t^{\text{dl}} + T_t^{\text{act}}. \quad (3)$$

Reducing radio time while increasing encoder or accelerator delay can worsen the loop. For a loop period h_i , jitter tolerance

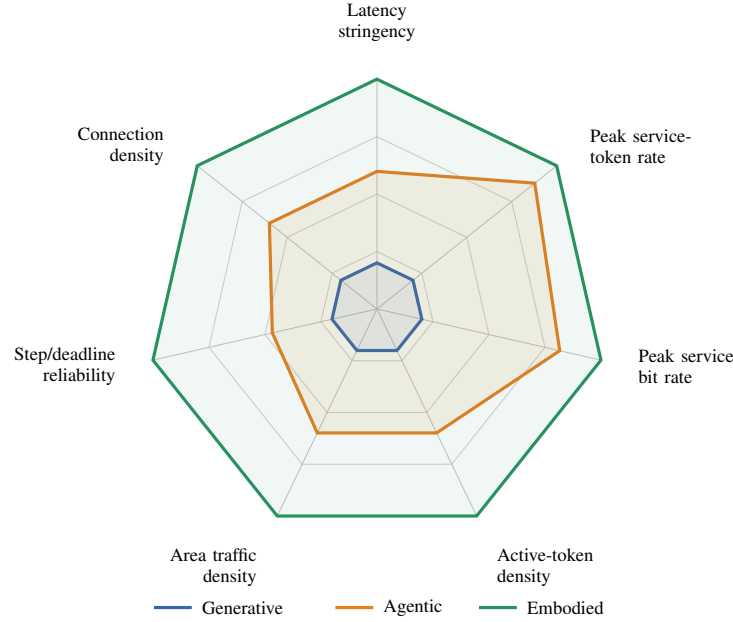


Fig. 3. Radar view of the seven QoS anchors in Table II. Each axis is logarithmically normalized between the smallest and largest tabulated anchor; the minimum is plotted at 0.2 rather than the origin for visibility. Latency and reliability increase outward as the latency budget and admissible failure probability decrease. The polygons compare illustrative scenario anchors, not standardized requirements or simultaneous maxima.

ϵ_i , and violation probability δ_i , a robust SLO is

$$\Pr\left\{T_{i,t}^{\text{loop}} \leq h_i, \left|T_{i,t}^{\text{loop}} - \mathbb{E}T_{i,t}^{\text{loop}}\right| \leq \epsilon_i h_i\right\} \geq 1 - \delta_i. \quad (4)$$

The first condition controls deadline reliability and the second jitter. Asynchronous modalities also require age and skew bounds: a perfectly delivered camera frame from before a contact transition can be less useful than a lower-resolution but current frame.

Together, the model metrics in Section II and the loop conditions above produce the most stringent numerical profile in Table II: a 1-ms illustrative network budget, 10^5 service tokens/s, up to 10 Gbit/s of aggregated multimodal traffic, and 99.999% deadline reliability. These values must be interpreted jointly. Higher sensing resolution and longer observation history can improve state estimation and task return, but increase encoding and inference time; faster action updates reduce reaction age, but repeat multimodal processing more frequently. An embodied service is therefore characterized not by a single peak rate, but by a coupled requirement on loop latency, tail jitter, cross-modal skew, deadline reliability, and control performance.

This coupling creates a latency–compute–performance burden for the network. High-rate sensor uplinks coexist with small, frequent action downlinks, and periodic control traffic is interrupted by event-triggered bursts near contact or environmental change. Lower source rates relieve radio and edge queues but can remove state detail; allocating more compute can improve perception or planning but may consume the same loop deadline that communication must meet. Short action packets also demand very low deadline-miss probability, whereas the redundancy needed to suppress rare failures consumes spectrum and path capacity. Dense deployments further synchronize sensing and action releases, so a small queueing

excursion can affect many loops at once. The resulting challenge is to sustain extreme tail reliability without allowing redundancy, multisource bursts, or accelerator contention to erase the latency and task performance gains it is intended to protect.

IV. FROM GENERATIVE TO EMBODIED NETWORKS FOR LAMS: ARCHITECTURE EVOLUTION

Architecture evolves along three coupled axes: *placement* across device, edge, and cloud; *network topology* among models, users, tools, and sensors; and *collaboration mode* among agents. Centralization is not a binary choice. One-to-one offloading, many-to-one aggregation, one-to-many dispatch, peer-to-peer exchange, and hierarchical hybrids coexist and should be selected by task dependency, latency, trust, and physical coupling.

A. Architecture for Generative-LAM Networks

A generative-LAM network is commonly cloud-centric. User equipment performs tokenization and lightweight preprocessing; an edge gateway provides admission, caching, privacy filtering, or small-model inference; and a cloud cluster hosts the full model. Model, tensor, pipeline, and expert parallelism distribute inference inside or across clusters. Wireless distributed mixture-of-experts systems additionally make expert routing dependent on channel and compute heterogeneity [44]. Pipeline parallelism and batching trade communication bubbles against accelerator utilization [45].

The dominant topology is one-to-one user–model access over a many-to-one serving hub. Users collaborate indirectly by sharing batches, model replicas, and KV-cache infrastructure, rather than by exchanging task state. The edge absorbs

latency-sensitive or private work and the cloud supplies model scale; this cloud–edge–device continuum is the basic mobile–edge–intelligence architecture for LLMs [22].

The main control loop is a slow resource-management loop: predict request and token lengths, place models, reserve accelerators and backhaul, batch compatible requests, and route overflow. Split inference places early layers on the device and later layers at the edge, transmitting intermediate features. Progressive feature transmission allows a receiver to stop once confidence is sufficient [46]. This architecture is effective when the user transaction is largely independent and deadlines are elastic. It becomes inefficient when every intermediate action depends on fresh tool or environment state.

B. Architecture for Agentic-LAM Networks

Agentic architecture adds a stateful orchestration plane. A planner or supervisor discovers capabilities, selects models and tools, maintains a task graph, and commits results to short- and long-term memory. Executors can be distributed over devices, edge sites, enterprise services, and clouds. Messages carry not only tokens but role, goal, schema, provenance, confidence, and dependency. Multi-agent systems use debate, vote, expert routing, or hierarchical decomposition; the best topology depends on task structure and the correlation of model errors [31], [47], [48]. One-to-many dispatch appears when a planner fans out to tools or specialists; many-to-one aggregation appears in voting and fusion; peer-to-peer exchange supports decentralized debate; and a supervisor plus peer subgraphs forms a hybrid. Emerging MCP, ACP, A2A, and ANP families cover different discovery, tool-access, messaging, and decentralization assumptions [35]. Thought- or latent-state exchange may further change the payload while retaining the same task graph [49], [50].

This design resembles a distributed transaction system but has probabilistic components. It requires:

- 1) capability and context discovery before routing;
- 2) session affinity for key–value caches and private memory;
- 3) checkpoint, retry, and idempotence semantics for tool calls;
- 4) verification paths for high-impact actions; and
- 5) joint compute–network–storage admission for the complete task rather than each RPC.

The architecture may keep a small model on-device for privacy and rapid screening, escalate uncertain inputs to a remote large model, and invoke specialists only when their expected information gain exceeds their latency and cost. Uncertainty-aware hybrid inference is one realization of this principle [51].

C. Architecture for Embodied-LAM Networks

Embodied systems require a hierarchy of loops (Fig. 4). A local reflex/control plane executes high-rate stabilization and safety interlocks. An edge cognition plane performs multi-modal fusion, VLA inference, local mapping, and coordination with deterministic access. A cloud learning plane provides large-model reasoning, fleet memory, simulation, and non-real-time model updates. A slow cloud model should not directly

occupy a hard real-time loop; it supplies goals, policies, latent plans, or verified skill updates to an edge/local controller. Decentralized local safety and reflexes coexist with temporarily centralized edge fusion. Consequently, collaboration becomes dynamic and partially centralized rather than globally centralized or fully peer-to-peer.

The architecture is also *partially centralized*. If two robots have neither task dependence nor physical interference, merging their observations in one model can add quadratic attention cost without useful information. When they jointly carry an object or occupy the same collision region, shared inference can improve consistency and reduce duplicated sensing. Model-sharing groups should therefore follow task dependencies, physical interactions, and complementary observations. Agents join a shared group when coordination is useful and separate when those dependencies disappear, rather than sending every observation to one global model. The resulting groups can run at different loop periods, producing an asynchronous, multi-rate networked control system.

V. IMPROVING NETWORKS FOR LAMS: KEY ENABLING TECHNOLOGIES

We group mechanisms into three interacting planes. The *token plane* decides what representation to compute and communicate. The *transport plane* decides how to deliver it by a deadline. The *coordination plane* decides which agents/models should share information and how network and compute resources close the loop. These planes are not a separate taxonomy: Table III maps each one to the common QoS scale in Table II.

A. Enablers for Generative-LAM Networks

1) *Compression and token communications*: Quantization, structured pruning, low-rank adaptation, key–value-cache compression, and top- k /top- p decoding reduce memory, computation, or output entropy. Token communications elevates tokens to semantic communication units and uses cross-modal context to reconstruct masked or corrupted content [26]. Compared with conventional source coding, a receiver equipped with a compatible foundation model can exploit learned priors. This can reduce payload but transfers burden to shared model state, tokenizer agreement, uncertainty calibration, and semantic error detection. For video generation, EVATok adapts sequence length to the content and exposes an explicit tradeoff between token count and reconstruction/generation quality [43]. Token-domain multiple access further lets many devices share tokenizer and modulation codebooks, using contextual reconstruction to mitigate token collisions [52]. It targets the high token and connection densities in Table II, but its gains depend on codebook compatibility and on whether reconstructed content satisfies the generative quality target.

The communication–quality tradeoff depends on the context available at the receiver. TokCom exploits cross-modal context for semantic reconstruction, while ToDMA uses contextual token recovery to mitigate collisions in shared access [26], [52]. For generative services, compression should therefore be evaluated against reconstructed content quality at a given

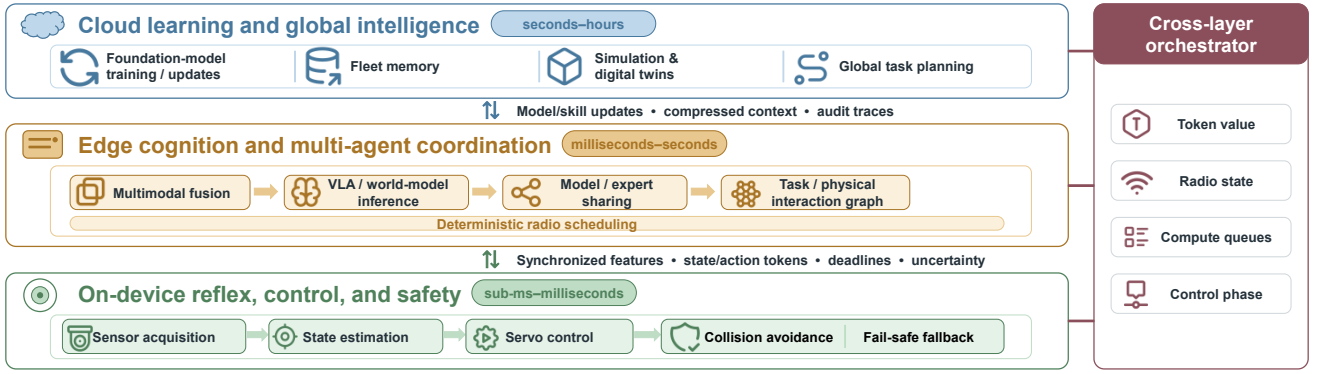


Fig. 4. Cloud-edge-device hierarchy for distributed LAM services. The generative stage primarily uses cloud scale, the agentic stage adds stateful edge orchestration, and the embodied stage keeps hard safety and stabilization local while the edge closes latency-critical cognition and coordination loops.

communication cost, rather than token reduction alone. Gains obtained with shared context should not be assumed to persist when that context is absent or inconsistent.

2) *Split, distributed, and speculative inference*: Split inference selects a cut layer and transmits intermediate features. Tensor parallelism distributes layer operations and repeatedly aggregates partial results; over-the-air computation can exploit wireless superposition to accelerate aggregation [53], [54]. Sparse-feature over-the-air fusion similarly combines multi-view features before reconstruction or inference [55]. These schemes are attractive when synchronized devices observe a common scene, but analog aggregation noise and stragglers must be evaluated at the task output. Intent-aware VLM split computing for disaster response is an early example of extending the placement decision toward embodied context [56].

Speculative decoding lets a small “draft” model propose tokens and a large model verify them, preserving the target model distribution while reducing serial decode cost [57]. Wireless implementations must jointly select the draft model, speculation length, offloading location, and radio resources. A low draft acceptance rate wastes both inference and communication. For generative services, the decision can be driven by expected decode speedup; for embodied services it must also account for action deviation and deadline risk [58].

B. Enablers for Agentic-LAM Networks

1) *Reliable collaboration and model routing*: Multi-model collaboration can improve coverage and verification through expert routing, debate, vote, or sequential refinement. The network problem is to choose the smallest collaboration graph that reaches a target task reliability before a deadline. Correlated models should not be treated as independent replicas: if they share training data or failure modes, majority vote offers less diversity than its communication cost suggests [31], [48].

In multi-agent workflows, passing the same task context between agents causes repeated transmission and prefill computation. Q-KVComm addresses this overhead by transmitting adaptively quantized KV caches [59], while KVComm shares selected informative KV layers [60]. Latent-space communication similarly avoids repeated conversion of intermediate rea-

soning into natural-language messages [50]. At a shared inference server, SemShareKV reuses cached computation across semantically similar prompts through token-level matching [61]. These mechanisms target the repeated context exchange and growing state of agentic tasks: representation sharing reduces inter-agent traffic, and cache reuse reduces redundant inference. Their use still requires compatible model state and application-controlled access to shared context.

Adaptive repeat querying provides a mechanism for reliable inference with heterogeneous edge LLMs [62]. More generally, an agent should stop when the expected value of another message is lower than its marginal compute, token, and latency cost. Network schedulers can use confidence and disagreement as hints, but final verification must remain application-controlled because confidence is not a safety certificate.

C. Enablers for Embodied-LAM Networks

1) *Control-loss-aware token selection*: To meet the sensing and loop-delay constraints in Section III, VLA acceleration reduces inference cost while preserving control-relevant information. FlashVLA selects visual tokens and reuses actions to reduce redundant inference [63], whereas QVLA allocates precision according to channel sensitivity measured at the action output [64]. DaDu-Corki jointly designs the embodied algorithm and accelerator architecture to reduce inference latency [65]. These results motivate evaluating compression by its effect on actions and task success, rather than representation fidelity alone. A remaining problem is to extend such empirical sensitivity measures to bounds on closed-loop control loss under dependent modalities and changing physical conditions.

2) *Finite-blocklength and deterministic transport*: Short action packets must meet the tight reliability and deadline targets in Section III. For blocklength n , signal-to-noise ratio γ , and error target ε , the finite-blocklength rate is approximated by

$$R(n, \varepsilon) \approx C(\gamma) - \sqrt{\frac{V(\gamma)}{n}} Q^{-1}(\varepsilon) + \frac{\log_2 n}{2n}, \quad (5)$$

where C and V are channel capacity and dispersion [66]. Finite-blocklength cell-free transmission and joint uplink/downlink allocation further expose hard-deadline and energy tradeoffs for multiple devices [67], [68]. This exposes the

bandwidth–latency–reliability tradeoff hidden by asymptotic capacity.

To sustain embodied-loop reliability, interface diversity, multi-connectivity, packet duplication, and optimized HARQ reduce outage or retransmission delay [69], [70]. Preemptive puncturing allows urgent traffic to overwrite scheduled broadband resources, while slicing isolates service classes [71]–[73]. Two-timescale QoS-aware multiplexing is useful when slow reservation must anticipate fast allocation under nonstationary demand [74]. Deterministic-delay work further optimizes jitter, hard-deadline violation, and high-order delay statistics under finite-blocklength transmission [75], [76].

However, coupled modalities and end-to-end deadlines limit direct transfer to embodied traffic. First, traditional models often assume independent sources and a fixed utility loss for punctured broadband packets. Multimodal embodied sources are coupled, so simultaneous degradation can remove all views of a critical object. Second, radio delay is only one term in (3). An end-to-end deterministic service must coordinate sensing release times, encoder and accelerator queues, radio slots, receiver inference, and actuation.

3) *Semantic and goal-oriented communication*: To relieve sensor-uplink congestion without losing control-relevant observations, goal-oriented communication prioritizes information by its task impact. Value and age of information have been applied to robotic waypoint delivery [77]. Integrated sensing and edge AI similarly co-designs sensing, communication, and inference around task performance [24]. For embodied LAMs, semantic communication should protect control objectives, cross-modal complementarity, and multiview complementarity. It should also report uncertainty: hallucinated recovery of a missing object is unacceptable when a safety controller interprets it as observation. Rate-limited closed-loop semantic communication offers a direct bridge from task representations to distributed sensing and control [78].

4) *Communication–control co-design and dynamic model sharing*: To address the coupled loop budget in Section III, wireless networked control jointly optimizes bandwidth, power, uplink/downlink ratio, sampling period, and control policy. Recent work considers sensing–communication–computing–control loops for robotic digital twins [79], communication–control codesign for large-scale systems [80], and communication allocation for system identification [81]. Energy-aware wireless communication–control co-design further shows that loop performance and radio efficiency should be optimized together [82]. These results show that optimizing physical control or identification can produce allocations very different from throughput maximization.

The dynamic groups in Section IV require joint control and resource scheduling. Relevant foundations include closed-loop allocation of uplink/downlink and edge resources [79], and communication–control scheduling for coupled loops [80]. Extending these foundations to embodied LAMs requires accounting for nonlinear learned policies, asynchronous sensing, and changing model-sharing groups. An important research direction is to coordinate token selection, model placement, loop periods, and radio/compute allocation while checking deadline reliability and physical task performance together.

VI. CASE STUDY: TWO-STAGE RECOGNITION OF LAM TRAFFIC

This case study connects the stage-specific traffic characteristics in Section III to practical service recognition. It first evaluates whether network-visible traffic alone distinguishes the three LAM classes, then outlines how explicit service descriptors can support QoS mapping.

LAM-aware QoS cannot assume that a mobile operator and an over-the-top (OTT) model provider already share service semantics. We therefore separate recognition into two deployment stages. In the *passive stage*, no such agreement exists. The operator observes packet timing, sizes, and directions and classifies each individual LAM traffic segment as generative, agentic, or embodied without using endpoint identifiers or payloads. Its decision may guide conservative QoS differentiation, but cannot certify a task deadline or infer private content. In the *active stage*, the operator and OTT provider agree on a small authenticated descriptor. Service identity, transaction dependency, deadline, and reliability bins can then be mapped to an admitted QoS profile without exposing prompts, reasoning traces, sensor content, or actions.

This separation makes the trust boundary explicit. Passive recognition asks whether LAM traffic has distinguishable time-domain signatures; below, measured traces and segment-classification results address this question. Active recognition asks how a declared class should be verified and sorted after commercial agreement; it is presented as a deployment framework rather than an experimental result.

A. Phase I: Passive Traffic Recognition Experiment

1) *Prototype and trace construction*: Fig. 5 shows the experimental platform. Two laptop terminals generate representative generative and agentic sessions, while a Uni-tree Go2 equipped with a Jetson edge module provides the embodied terminal. All terminals traverse the same access gateway. A monitor port supplies packet timestamps, sizes, and directions for trace construction and offline classification; packet payloads are not inspected. The workstation hosts the large-model inference services: Qwen3-8B [84] for text Q&A and Janus-Pro-7B [85], [86] for image understanding and image generation. For agentic workloads, we deploy OpenClaw with Qwen3-8B for file creation, file moving, and code writing; for embodied navigation, we deploy InternVLA-N1-DualVLN [87]. To construct the traces, we aggregate captured packet sizes separately for uplink and downlink into 20-ms bins and report the traffic volume in kbit per bin, with timestamps measured relative to the start of each workload.

2) *Time-domain payload characteristics*: Fig. 6 presents the measured traffic traces. *Generative* traffic depends on the input and output modalities: text Q&A comprises a short request upload followed by a streamed text response; image understanding requires a larger image upload before the text response; image generation exhibits a long interval with little traffic before a concentrated image download. *Agentic* file creation, file moving, and code writing follow a multi-round sequence: context upload, a streamed tool-call response, local tool execution, a second upload containing context and tool

TABLE III
STAGE-BY-STAGE MAPPING FROM QoS AND ARCHITECTURE TO ENABLING TECHNOLOGIES

Stage	QoS/model target	Placement/topology	Token/data-plane enablers	Transport/compute enablers	Coordination enablers and open gap
Generative	100-ms/ 10^3 -token/s anchor; content fidelity, useful-token goodput, and cost	Cloud-dominant one-to-one access over many-to-one edge/cloud serving	Adaptive tokenization; token pruning/quantization; TokCom and ToDMA [26], [52]	Batch/cache-aware scheduling; split/progressive inference; speculative decode [46], [57]	Joint radio-compute admission and model/expert placement; open gap is generalization across modalities, tokenizers, and model versions
Agentic	10-ms/5 × 10^4 -token/s/99.9% anchors; task goodput, memory freshness, job tail, and cost	One-to-many tools, many-to-one aggregation, peer debate, or hybrid task graph	KV quantization/selection, semantic cache reuse, machine/latent-state exchange [50], [59]–[61]	Transactional partial reliability; cache affinity; early exit, heterogeneous retry, and joint network-compute-storage admission	Network-aware tool routing and adaptive repeat querying [62], [83]; open gap is reliable interoperability without exposing private plans or memory
Embodied	1-ms/ 10^5 -token/s/99.999% anchors; loop tail, jitter, synchronized goodput, control cost, and safety	Local reflex plus temporary edge-centered clusters on a dynamic many-to-many interaction graph	Control-loss-aware token/channel selection, task-conditioned compression, semantic/goal-oriented updates	Finite-blocklength coding, diversity/HARQ, slicing, deterministic radio-compute scheduling, and safe local fallback	Interaction-graph clustering, multi-rate model sharing, and communication-control co-design [79], [80]; open gap is a bound from nonlinear token error to long-horizon physical loss

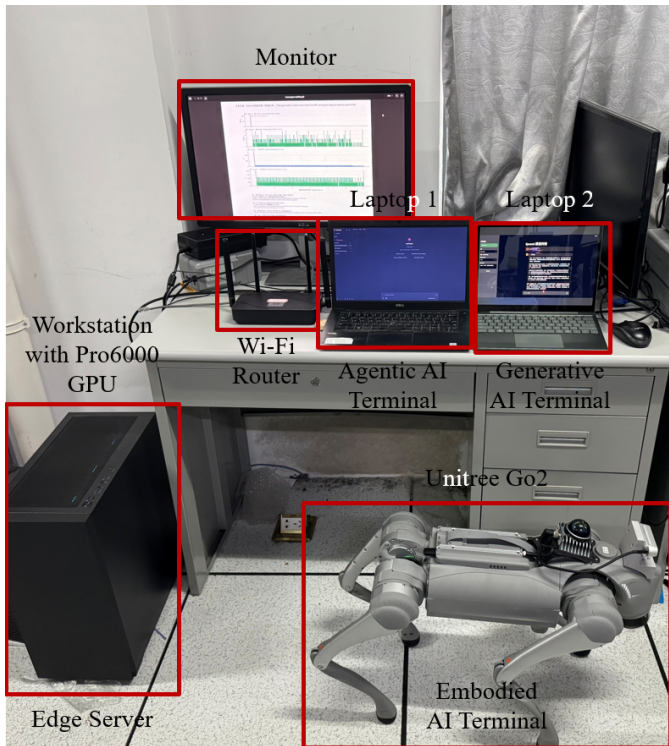


Fig. 5. Experimental platform for the Phase-I passive traffic-recognition experiment. Laptop 1 and Laptop 2 generate agentic- and generative-AI traffic, respectively, while the Unitree Go2 serves as the embodied-AI terminal. The three terminals connect through a common Wi-Fi router. A workstation/edge server equipped with a Pro 6000 GPU hosts large models and provides inference services to the terminals. The monitor displays the resulting up-link/downlink traffic traces for passive-recognition analysis.

results, and a final reply. The file-moving trace makes this traffic-silent execution interval particularly visible. *Embodied* navigation exhibits recurrent large RGB-frame uplink bursts interleaved with much smaller action/trajectory downlinks throughout the task, reflecting repeated perception-action exchanges. These measurements corroborate the traffic characteristics discussed in Section III for the tested workloads: modality-dependent asymmetry and burstiness, task-driven intermittent exchanges, and sustained sensor-action coupling, respectively.

TABLE IV
BLIND THREE-CLASS SEGMENT-CLASSIFICATION TEST RESULTS (%)

Method	Accuracy	F1-G	F1-A	F1-E	Macro-F1
Statistical linear	99.78	99.75	99.75	100.00	99.83
1D-CNN	100.00	100.00	100.00	100.00	100.00
BiLSTM	99.78	99.75	99.75	100.00	99.83

G/A/E denote generative/agentic/embodied LAMs. Test support is 200/200/60 segments, respectively.

3) *Blind segment-classification protocol*: The dataset contains 2,300 individual 10s traffic segments: 1,000 generative (G), 1,000 agentic (A), and 300 embodied (E) segments. A fixed 70%/10%/20% split yields 1,610 training, 230 validation, and 460 test samples; the test set contains 200 generative, 200 agentic, and 60 embodied samples. Each segment is aggregated into 500 bins of 20 ms, yielding

$$\mathbf{x}_t = [B_t^{\text{UL}}, B_t^{\text{DL}}, N_t^{\text{UL}}, N_t^{\text{DL}}], \quad \mathbf{X} \in \mathbb{R}^{500 \times 4}, \quad (6)$$

where B_t and N_t denote frame-byte and packet counts, respectively. Only these network-observable dynamics are used; payloads, endpoint and application identifiers, prompts, tokens, task categories, and outcomes are excluded from classifier inputs.

4) *Baselines and experimental results*: We compare a statistical linear baseline [88], a 1D-CNN [89], and a bidirectional LSTM (BiLSTM) [90], adapting these classifier families to the same four-channel observations. Macro-F1 is the primary metric to account for class imbalance. Before test evaluation, validation Macro-F1 with a simplicity tie-break selects the statistical linear baseline as the primary model.

Table IV reports the test results. The statistical linear baseline and BiLSTM both achieve 99.78% accuracy and 99.83% Macro-F1, while the 1D-CNN correctly classifies all 460 test segments. Fig. 7 shows that the primary model makes only one error, labeling a generative segment as agentic; all 60 embodied segments are correctly identified. Post-hoc diagnostic ablations retain 100% test Macro-F1 for volume-normalized and packet-only CNN inputs, while a linear baseline without absolute aggregates achieves 99.14% Macro-F1.

These results demonstrate distinctive traffic characteristics across LAM stages, enabling effective blind recognition without payload inspection or application identifiers under the

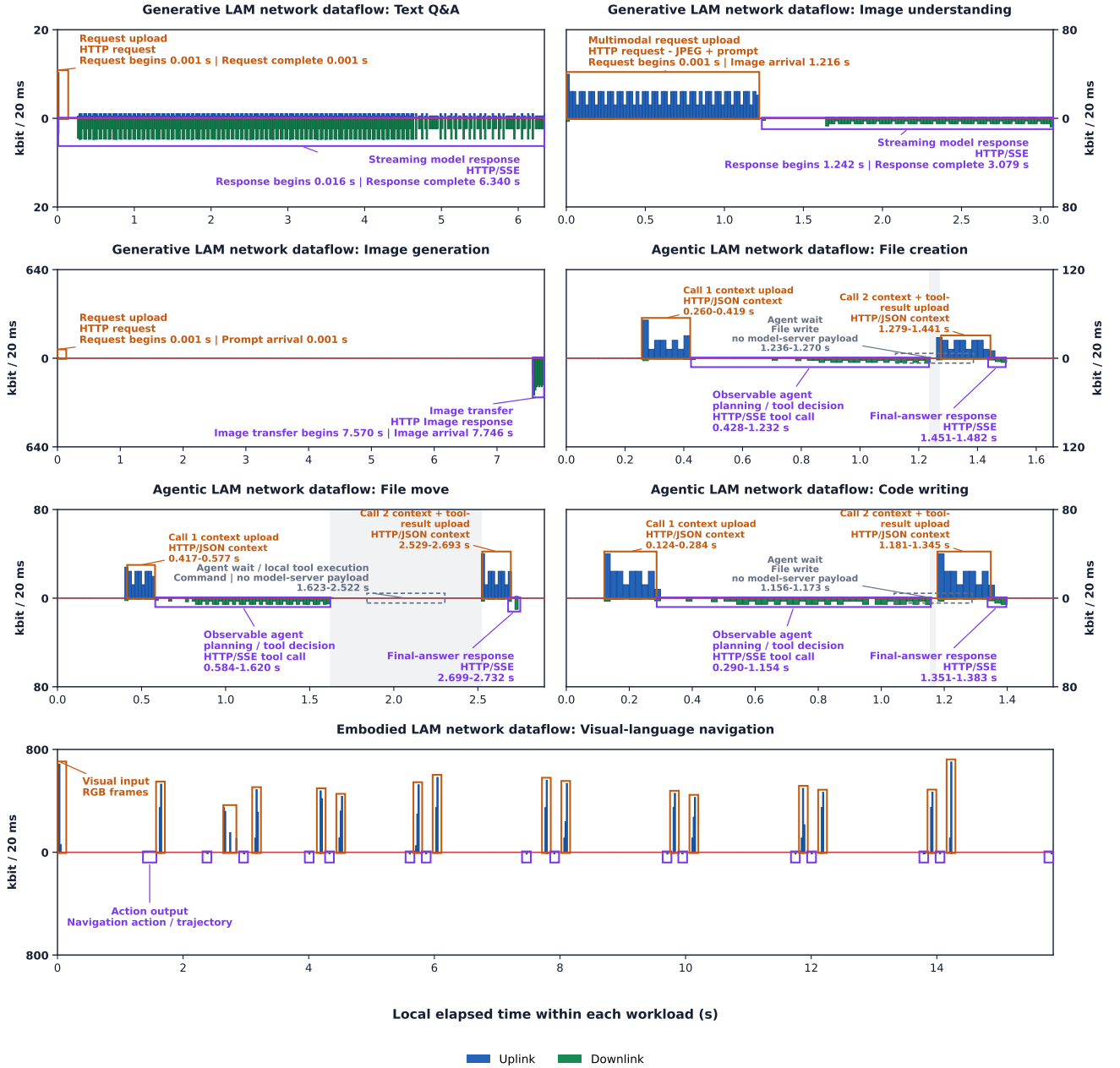


Fig. 6. Measured traffic traces for three generative workloads (text Q&A, image understanding, and image generation), three agentic workloads (file creation, file moving, and code writing), and embodied vision-and-language navigation. Blue and green bars show uplink and downlink traffic volumes in 20-ms bins, respectively.

evaluated capture conditions. generalization remains to be verified.

B. Phase II: Active Recognition and QoS Mapping

Active recognition augments traffic observations with a compact authenticated descriptor whenever the application provider makes such information available. As illustrated in Fig. 8, the descriptor carries an opaque service ID, declared generative, agentic, embodied, or unknown class, session/chunk identifier, timestamp, deadline bin, reliability tier, and token- or byte-count bin. These fields let the network associate a declared service with an admitted QoS profile without inferring application semantics from encrypted payloads.

Passive traffic features can remain available as coarse consistency and monitoring signals, so the two recognition modes can coexist rather than requiring a rigid switch. Descriptor validation checks its signature, authorization, freshness, and binding to the admitted session. A substantial traffic-shape mismatch may prompt verification, but does not silently override an authenticated declaration.

The framework operates at three granularities. *Registration-time recognition* verifies the service manifest, tenant authorization, endpoint attestation, and maximum admitted rate/reliability class. *Session-level recognition* binds the declaration to observed traffic and to model, memory, or edge-compute affinity. *Chunk-level sorting* uses endpoint-assigned

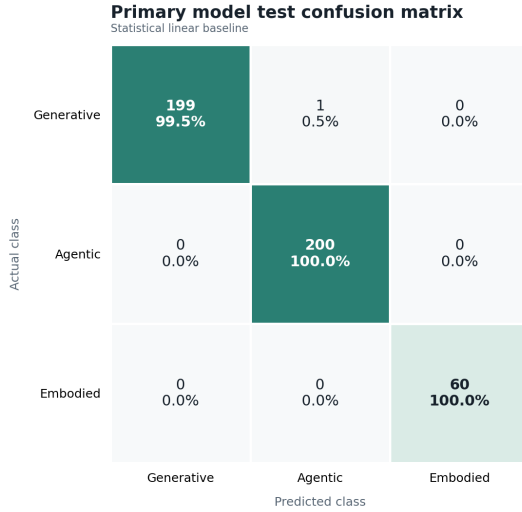


Fig. 7. Test confusion matrix of the validation-selected statistical linear baseline. ($n = 460$).

deadline, dependency, and coarse value bins to order traffic inside the admitted envelope; the network does not infer semantic value from encrypted payloads.

QoS mapping follows Section III and Table II. Generative sessions map directional bit/token rates and burst demand to uplink/downlink resource allocations. Agentic sessions additionally use task dependencies and expiry information to support per-step reliability, state freshness, and task-completion time. Embodied sessions use sensing/action timestamps to enforce loop latency, jitter, cross-modal skew, and deadline reliability. The tabulated numerical anchors remain scenario-dependent rather than fixed guarantees. If metadata is unavailable, passive recognition can continue to provide traffic-level coverage, whereas valid descriptors enable auditable QoS mapping without requiring the operator to inspect model inputs, private memory, or physical task semantics.

VII. CHALLENGES AND FUTURE DIRECTIONS

A. Tokenizer Compatibility

Distributed LAM services increasingly exchange intermediate observations, context, and reasoning state between agents, rather than only delivering a final response to a user. Tokens offer a model-facing representation for transmitting and fusing this information without repeatedly reconstructing raw multi-modal inputs. Researches in TokCom exploits cross-modal context for semantic reconstruction [26], and their ToDMA extends token-domain transmission to many devices sharing wireless resources [52].

Token communications assumes that sender and receiver agree on token boundaries, codebooks, embeddings, and model semantics. In practice, tokenizers differ by model, version, language, modality, and embodiment. Even identical token IDs are meaningless without a versioned vocabulary and embedding interpretation. Visual and action tokenizers are especially unstable because codebooks can change during fine-tuning.

Compatibility has four layers:

- 1) *syntactic*: vocabulary and token-ID mapping;
- 2) *representational*: embedding dimension, quantization, and feature normalization;
- 3) *semantic*: equivalent meaning or task effect across models; and
- 4) *behavioral*: equivalent downstream action within a safety tolerance.

Only the first two can be solved by a schema registry. Semantic and behavioral compatibility require calibration data, adapters, or a shared latent interface. A universal tokenizer may simplify networking but can sacrifice model efficiency and lock the ecosystem to one representation.

Future protocols should negotiate a tokenizer/model manifest containing a cryptographic version hash, modality, vocabulary or codebook identifier, embedding format, supported adapters, calibration range, and fallback codec. Gateways can translate tokens through learned adapters, but translation adds latency and the translation error must be considered. For safety-critical actions, a receiver should reject an unknown or out-of-range codebook rather than guess.

Model updates create a consistency problem similar to a rolling protocol upgrade. During a fleet update, old and new tokenizers may coexist. Dual encoding is robust but expensive; atomic migration is operationally hard. Version-aware routing and shadow validation can support a staged transition. A major research need is a formal bound connecting embedding alignment error to task and control degradation.

B. Token-Native Protocols

Token-native protocols should add transaction and token/chunk metadata over existing packet transports, while remaining usable by token-unaware networks. Consistent with Section VI, rich model/tokenizer, modality, dependency, and uncertainty records stay at endpoints; operators receive only an opaque service/session handle, size, expiry, and authenticated reliability class. These fields can support deadline-aware scheduling and selective recovery without exposing payload semantics.

Congestion feedback should account for radio load, admitted edge-compute capacity, and deadline-miss risk, so expired chunks and overloaded inference queues do not consume resources needed by other services. Such metadata also creates privacy and integrity risks: forged priorities, stale actions, or incompatible tokenizer versions can invalidate a task. Authentication, version checks, replay protection, and application-side authorization must therefore accompany scheduling, with enforced isolation for conventional and nonparticipating traffic.

C. Benchmarking, Theory, and Standardization Gaps

Current benchmarks usually isolate model accuracy, network traces, or control dynamics. The community needs co-simulation and hardware testbeds that combine real model serving, wireless queues, clock drift, sensor release patterns, and physical episodes. Benchmarks should span generative, agentic, and embodied traffic on one infrastructure so that resource competition and backward compatibility can be studied. Shared datasets should include task/physical interaction graphs

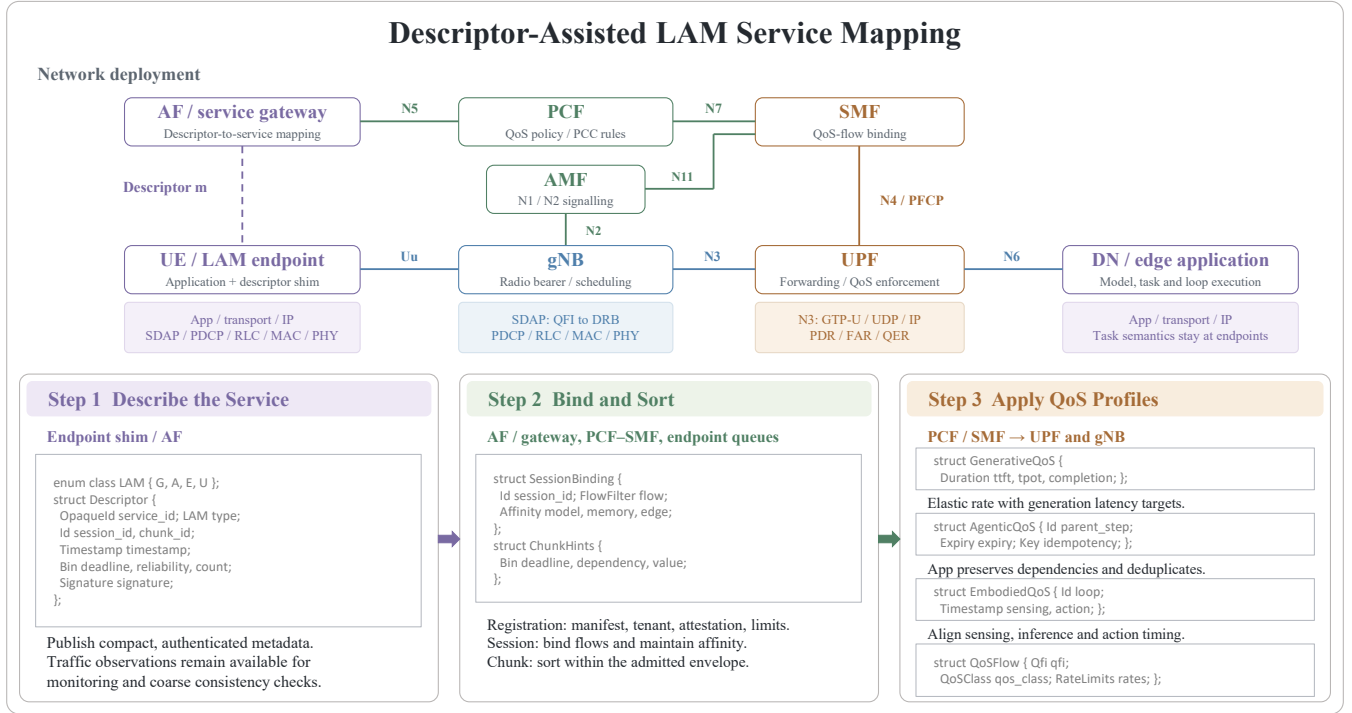


Fig. 8. Illustration of functional units and procedures tailored for active descriptor-assisted LAM service mapping.

and time-aligned token-importance labels, while protecting private sensory content.

Theory must move beyond independent sources, stationary delay, linear plants, and known dynamics. Promising tools include multi-terminal task-rate-distortion theory, risk-sensitive and distributionally robust optimization, event-triggered control, graph sparsification, and online learning with safety shields. The central missing bound connects four errors: token representation distortion, wireless delivery uncertainty, model prediction error, and long-horizon physical control loss.

Standardization should proceed incrementally. Near-term systems can define LAM service descriptors and telemetry over existing QoS flows. A second step can standardize capability/tokenizer manifests and compute-aware admission. Only after interoperability and security experience should fine-grained token-native transport be exposed across domains. The active 3GPP 6G use-case study [14] provides a venue, but task semantics and safety policy must remain under application and vertical-domain governance.

VIII. CONCLUSIONS

This survey traces the evolution from generative content delivery to agentic task execution and embodied closed-loop interaction. Their distinctive network demands are, respectively, asymmetric and bursty multimodal traffic, stateful multi-round exchanges with reliability and freshness requirements, and tightly coupled sensing-inference-action traffic with stringent delay, jitter, and deadline reliability. Corresponding architectures evolve from cloud-centric serving to distributed task graphs and local-edge-cloud coordination. Efficient inference and token compression, reusable agent state and reliable tool

interactions, and communication-control co-design address these different bottlenecks; the numerical QoS anchors remain illustrative deployment scenarios rather than universal requirements.

The case study provides measurement-based evidence for these traffic characteristics. On 2,300 individual 10-s segments, blind recognition achieves 99.78% test accuracy with the statistical linear baseline and BiLSTM, and 100% with the 1D-CNN, without payload or application identifiers. The results establish separability under the evaluated capture conditions, while cross-environment generalization remains open. Complementing passive classification, descriptor-assisted recognition outlines how declared service requirements can map to network QoS without exposing private semantics; it is a deployment framework, not an experimentally validated mechanism. Tokenizer interoperability, secure token-aware protocols, and joint network-model-control evaluation remain priorities for connecting distributed LAM intelligence to physical systems.

REFERENCES

- [1] NVIDIA, “What is generative AI?” Accessed: Oct. 1, 2026. [Online]. Available: <https://www.nvidia.com/en-us/glossary/generative-ai/>
- [2] T. B. Brown *et al.*, “Language models are few-shot learners,” in *Proc. NeurIPS*, vol. 33, 2020, pp. 1877–1901.
- [3] A. Ramesh *et al.*, “Zero-shot text-to-image generation,” in *Proc. ICML*, 2021, pp. 8821–8831.
- [4] D. Hendrycks *et al.*, “Measuring massive multitask language understanding,” in *Proc. ICLR*, 2021.
- [5] A. Srivastava *et al.*, “Beyond the imitation game: Quantifying and extrapolating the capabilities of language models,” *Trans. Mach. Learn. Res.*, May 2023.
- [6] X. Liu *et al.*, “AgentBench: Evaluating LLMs as agents,” in *Proc. ICLR*, 2024.

- [7] S. Yao *et al.*, “ReAct: Synergizing reasoning and acting in language models,” in *Proc. ICLR*, 2023.
- [8] N. Shinn, F. Cassano, A. Gopinath, K. Narasimhan, and S. Yao, “Reflexion: Language agents with verbal reinforcement learning,” in *Proc. NeurIPS*, vol. 36, 2023, pp. 8634–8652.
- [9] T. Yu *et al.*, “Meta-World: A benchmark and evaluation for multi-task and meta reinforcement learning,” in *Proc. Conf. Robot Learn.*, 2020, pp. 1094–1100.
- [10] B. Zitkovich *et al.*, “RT-2: Vision-language-action models transfer web knowledge to robotic control,” in *Proc. Conf. Robot Learn.*, 2023, pp. 2165–2183.
- [11] B. Liu *et al.*, “LIBERO: Benchmarking knowledge transfer for lifelong robot learning,” in *Proc. NeurIPS*, vol. 36, 2023, pp. 44776–44791.
- [12] Ericsson, “The rise of uplink demand in AI-driven mobile networks,” *Ericsson Mobility Report*, Jun. 2026. Accessed: Oct. 1, 2026. [Online]. Available: <https://www.ericsson.com/en/reports-and-papers/mobility-report/articles/rising-uplink-demand-ai-driven-mobile-networks>
- [13] ITU-R, “Framework and overall objectives of the future development of IMT for 2030 and beyond,” Recommendation ITU-R M.2160-0, Nov. 2023.
- [14] 3GPP, “Study on 6G use cases and service requirements,” 3rd Generation Partnership Project (3GPP), Tech. Rep. 22.870, ver. 20.0.0, Rel. 20, Mar. 2026.
- [15] European Commission, “Hexa-X-II,” CORDIS project 101095759, 2023–2025. Accessed: Oct. 1, 2026. [Online]. Available: <https://cordis.europa.eu/project/id/101095759>
- [16] European Commission, “6G Goal-Oriented AI-enabled Learning and Semantic Communication Networks (6G-GOALS),” CORDIS project 101139232, 2024–2026. Accessed: Oct. 1, 2026. [Online]. Available: <https://cordis.europa.eu/project/id/101139232>
- [17] AI-RAN Alliance, “AI-RAN Alliance reaches major milestone, showcasing breakthrough momentum for AI-native networks,” Feb. 26, 2026. Accessed: Oct. 1, 2026. [Online]. Available: <https://ai-ran.org/press-releases/mwc-2026-momentum>
- [18] The Linux Foundation, “Linux Foundation launches the Agent2Agent protocol project to enable secure, intelligent communication between AI agents,” Jun. 23, 2025. Accessed: Oct. 1, 2026. [Online]. Available: <https://www.linuxfoundation.org/press/linux-foundation-launches-the-agent2agent-protocol-project-to-enable-secure-intelligent-communication-between-ai-agents>
- [19] J. Björck *et al.*, “GR00T N1: An open foundation model for generalist humanoid robots,” arXiv:2503.14734, Mar. 2025.
- [20] Z. Xi *et al.*, “The rise and potential of large language model based agents: A survey,” *Sci. China Inf. Sci.*, vol. 68, no. 2, Art. no. 121101, 2025, doi: 10.1007/s11432-024-4222-0.
- [21] Y. Liu *et al.*, “Aligning cyber space with physical world: A comprehensive survey on embodied AI,” *IEEE/ASME Trans. Mechatronics*, vol. 30, no. 6, pp. 7253–7274, Dec. 2025, doi: 10.1109/TMECH.2025.3574943.
- [22] G. Qu, Q. Chen, W. Wei, Z. Lin, X. Chen, and K. Huang, “Mobile edge intelligence for large language models: A contemporary survey,” *IEEE Commun. Surveys Tuts.*, vol. 27, no. 6, pp. 3820–3860, Dec. 2025, doi: 10.1109/COMST.2025.3527641.
- [23] S. Guo *et al.*, “A survey on semantic communication networks: Architecture, security, and privacy,” *IEEE Commun. Surveys Tuts.*, vol. 27, no. 5, pp. 2860–2894, Oct. 2025, doi: 10.1109/COMST.2024.3516819.
- [24] Z. Liu *et al.*, “Integrated sensing and edge AI: Realizing intelligent perception in 6G,” *IEEE Commun. Surveys Tuts.*, vol. 28, pp. 2725–2770, 2026, doi: 10.1109/COMST.2025.3592989.
- [25] H. Xie, Z. Qin, G. Y. Li, and B.-H. Juang, “Deep learning enabled semantic communication systems,” *IEEE Trans. Signal Process.*, vol. 69, pp. 2663–2675, 2021.
- [26] L. Qiao, M. B. Mashhadi, Z. Gao, R. Tafazolli, M. Bennis, and D. Niyato, “Token communications: A large model-driven framework for cross-modal context-aware semantic communications,” *IEEE Wireless Commun.*, vol. 32, no. 5, pp. 80–88, Oct. 2025.
- [27] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, “BLEU: A method for automatic evaluation of machine translation,” in *Proc. ACL*, 2002, pp. 311–318.
- [28] C.-Y. Lin, “ROUGE: A package for automatic evaluation of summaries,” in *Text Summarization Branches Out*, 2004, pp. 74–81.
- [29] A. Dosovitskiy *et al.*, “An image is worth 16×16 words: Transformers for image recognition at scale,” in *Proc. ICLR*, 2021.
- [30] J. Ho, A. Jain, and P. Abbeel, “Denoising diffusion probabilistic models,” in *Proc. NeurIPS*, vol. 33, 2020, pp. 6840–6851.
- [31] Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, and I. Mordatch, “Improving factuality and reasoning in language models through multiagent debate,” in *Proc. ICML*, 2024, pp. 11733–11763.
- [32] J. Wei *et al.*, “Chain-of-thought prompting elicits reasoning in large language models,” in *Proc. NeurIPS*, vol. 35, 2022, pp. 24824–24837.
- [33] S. Yao *et al.*, “Tree of thoughts: Deliberate problem solving with large language models,” in *Proc. NeurIPS*, vol. 36, 2023, pp. 11809–11822.
- [34] P. Lewis *et al.*, “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in *Proc. NeurIPS*, vol. 33, 2020, pp. 9459–9474.
- [35] A. Ehteshami, A. Singh, G. K. Gupta, and S. Kumar, “A survey of agent interoperability protocols: Model Context Protocol (MCP), Agent Communication Protocol (ACP), Agent-to-Agent Protocol (A2A), and Agent Network Protocol (ANP),” arXiv:2505.02279, May 2025.
- [36] S. Zhou *et al.*, “WebArena: A realistic web environment for building autonomous agents,” in *Proc. ICLR*, 2024.
- [37] C. E. Jimenez *et al.*, “SWE-bench: Can language models resolve real-world GitHub issues?” in *Proc. ICLR*, 2024.
- [38] D. Driess *et al.*, “PaLM-E: An embodied multimodal language model,” in *Proc. ICML*, 2023, pp. 8469–8488.
- [39] D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap, “Mastering diverse control tasks through world models,” *Nature*, vol. 640, pp. 647–653, Apr. 2025, doi: 10.1038/s41586-025-08744-2.
- [40] C. Sferazza, D.-M. Huang, X. Lin, Y. Lee, and P. Abbeel, “Humanoid-Bench: Simulated humanoid benchmark for whole-body locomotion and manipulation,” in *Proc. Robot.: Sci. Syst. (RSS)*, Jul. 2024, doi: 10.15607/RSS.2024.XX.061.
- [41] X. Puig *et al.*, “Habitat 3.0: A co-habitat for humans, avatars, and robots,” in *Proc. ICLR*, 2024.
- [42] Y. Chen, W. Zhang, and X. Li, “Reflex: Enabling fast and predictive vision-language-action models for reaction-critical manipulation,” arXiv:2608.14379, Aug. 2026.
- [43] T. Xiong, J. H. Liew, Z. Huang, Z. Lin, J. Feng, and X. Liu, “EVAtoK: Adaptive length video tokenization for efficient visual autoregressive generation,” in *Proc. IEEE/CVF CVPR*, Jun. 2026, pp. 23249–23259.
- [44] N. Xue *et al.*, “WDMoE: Wireless distributed mixture of experts for large language models,” *IEEE Trans. Wireless Commun.*, vol. 25, pp. 559–572, 2026.
- [45] J. Jiang, Z. Chen, H. Shin, and A. Nallanathan, “Edge inference for large language models with pipeline parallelism and batching,” *IEEE Trans. Commun.*, vol. 74, pp. 8390–8406, 2026, doi: 10.1109/TCOMM.2026.3686718.
- [46] Q. Lan, Q. Zeng, P. Popovski, D. Gündüz, and K. Huang, “Progressive feature transmission for split classification at the wireless edge,” *IEEE Trans. Wireless Commun.*, vol. 22, no. 6, pp. 3837–3852, Jun. 2023.
- [47] X. Zheng *et al.*, “Exploring multi-agent dynamics for generative AI and large language models in mobile edge networks,” *IEEE Wireless Commun.*, vol. 32, no. 6, pp. 69–78, Dec. 2025.
- [48] A. Estornell and Y. Liu, “Multi-LLM debate: Framework, principals, and interventions,” in *Proc. NeurIPS*, vol. 37, 2024, pp. 28938–28964.
- [49] Y. Zheng *et al.*, “Thought communication in multiagent collaboration,” in *Proc. NeurIPS*, vol. 38, 2025, pp. 123389–123418.
- [50] Z. Du *et al.*, “Enabling agents to communicate entirely in latent space,” in *Proc. ACL*, 2026, pp. 27106–27129.
- [51] S. Oh, J. Kim, J. Park, S.-W. Ko, T. Q. S. Quek, and S.-L. Kim, “Uncertainty-aware hybrid inference with on-device small and remote large language models,” in *Proc. IEEE ICMLCN*, 2025, pp. 1–7.
- [52] L. Qiao, M. B. Mashhadi, Z. Gao, R. Schober, and D. Gündüz, “ToDMA: Large model-driven massive token communications for semantic multiple access,” arXiv:2505.10946v3, Jul. 2026.
- [53] K. Zhang, H. He, S. Song, J. Zhang, and K. B. Letaief, “Communication-efficient distributed on-device LLM inference over wireless networks,” *IEEE J. Sel. Topics Signal Process.*, vol. 19, no. 7, pp. 1301–1317, Oct. 2025.
- [54] Y. Cang, M. Chen, and K. Huang, “Over-the-air computation for realizing neural link in in-network AI architectures,” *IEEE Trans. Wireless Commun.*, vol. 25, pp. 2373–2388, 2026.
- [55] Z. Liu, Q. Lan, and K. Huang, “Over-the-air fusion of sparse spatial features for integrated sensing and edge AI over broadband channels,” *IEEE Trans. Wireless Commun.*, vol. 24, no. 4, pp. 2999–3013, Apr. 2025.
- [56] R. Bhattacharjya *et al.*, “AVERY: Intent-driven adaptive VLM split computing via embodied self-awareness for efficient disaster response systems,” arXiv:2511.18151v3, Mar. 2026.
- [57] Y. Leviathan, M. Kalman, and Y. Matias, “Fast inference from transformers via speculative decoding,” in *Proc. ICML*, 2023, pp. 19274–19286.
- [58] J. Park, Y. Lim, S. Oh, J. Park, J. Choi, and S.-L. Kim, “Action deviation-aware inference for low-latency wireless robots,” arXiv:2510.02851v2, Nov. 2025.
- [59] B. Kriuk and L. Ng, “Q-KVComm: Efficient multi-agent communication via adaptive KV cache compression,” arXiv:2512.17914, Nov. 2025.

- [60] X. Shi, M. Chiesa, G. Q. Maguire Jr., and D. Kostić, “KVComm: Enabling efficient LLM communication through selective KV sharing,” in *Proc. ICLR*, 2026.
- [61] X. Zhao and S. Mastorakis, “SemShareKV: Efficient KVCache sharing for semantically similar prompts via token-level LSH matching,” in *Findings Assoc. Comput. Linguistics: IJCNLP-AACL*, Dec. 2025, pp. 432–445, doi: 10.18653/v1/2025.findings-ijcnlp.25.
- [62] T. Wang *et al.*, “Edge heterogeneous LLMs collaborative reliable inference based on adaptive repeat query,” in *Proc. IEEE ICC Workshops*, 2026, pp. 1–6, doi: 10.1109/ICCSWorkshops63917.2026.11586445.
- [63] X. Tan *et al.*, “Think twice, act once: Token-aware compression and action reuse for efficient inference in vision-language-action models,” arXiv:2505.21200, May 2025.
- [64] Y. Xu *et al.*, “QVLA: Not all channels are equal in vision-language-action model’s quantization,” in *Proc. ICLR*, 2026.
- [65] Y. Huang *et al.*, “DaDu-Corki: Algorithm-architecture co-design for embodied AI-powered robotic manipulation,” in *Proc. ACM/IEEE ISCA*, 2025, pp. 327–343.
- [66] Y. Polyanskiy, H. V. Poor, and S. Verdú, “Channel coding rate in the finite blocklength regime,” *IEEE Trans. Inf. Theory*, vol. 56, no. 5, pp. 2307–2359, May 2010.
- [67] Z. Zhang *et al.*, “Performance of multidevice downlink cell-free system under finite blocklength for URLLC with hard deadlines,” *IEEE J. Sel. Areas Commun.*, vol. 41, no. 7, pp. 2090–2106, Jul. 2023.
- [68] K. Li, P. Zhu, Y. Wang, F.-C. Zheng, and X. You, “Joint uplink and downlink resource allocation toward energy-efficient transmission for URLLC,” *IEEE J. Sel. Areas Commun.*, vol. 41, no. 7, pp. 2176–2192, Jul. 2023.
- [69] J. J. Nielsen, R. Liu, and P. Popovski, “Ultra-reliable low latency communication using interface diversity,” *IEEE Trans. Commun.*, vol. 66, no. 3, pp. 1322–1334, Mar. 2018.
- [70] A. Anand and G. de Veciana, “Resource allocation and HARQ optimization for URLLC traffic in 5G wireless networks,” *IEEE J. Sel. Areas Commun.*, vol. 36, no. 11, pp. 2411–2421, Nov. 2018.
- [71] A. Anand, G. de Veciana, and S. Shakkottai, “Joint scheduling of URLLC and eMBB traffic in 5G wireless networks,” *IEEE/ACM Trans. Netw.*, vol. 28, no. 2, pp. 477–490, Apr. 2020.
- [72] P. Yang, X. Xi, T. Q. S. Quek, J. Chen, X. Cao, and D. Wu, “RAN slicing for massive IoT and bursty URLLC service multiplexing: Analysis and optimization,” *IEEE Internet Things J.*, vol. 8, no. 18, pp. 14258–14275, Sep. 2021.
- [73] M. Alsenwi, N. H. Tran, M. Bennis, S. R. Pandey, A. K. Bairagi, and C. S. Hong, “Intelligent resource slicing for eMBB and URLLC coexistence in 5G and beyond: A deep reinforcement learning based approach,” *IEEE Trans. Wireless Commun.*, vol. 20, no. 7, pp. 4585–4600, Jul. 2021.
- [74] J. Li *et al.*, “A unified QoS-aware multiplexing framework for next-generation immersive communication with legacy wireless applications,” *IEEE Internet Things J.*, vol. 12, no. 14, pp. 28582–28597, Jul. 2025.
- [75] Z. Liu, K. Li, Y. Wang, and P. Zhu, “Mixed traffic scheduling with latency and jitter analysis in URLLC industrial automation,” *IEEE Internet Things J.*, vol. 12, no. 12, pp. 18820–18835, Jun. 2025.
- [76] H. Ding, S. Xu, Z. Xu, R. Xu, and B. Ai, “Achieving deterministic delay in industrial Internet of Things: Resource allocation schemes in the finite blocklength regime,” *IEEE Trans. Veh. Technol.*, vol. 74, no. 9, pp. 14192–14206, Sep. 2025.
- [77] W. Wu, Y. Yang, Y. Deng, and A. H. Aghvami, “Goal-oriented semantic communications for robotic waypoint transmission: The value and age of information approach,” *IEEE Trans. Wireless Commun.*, vol. 23, no. 12, pp. 18903–18915, Dec. 2024.
- [78] G. Pan, A. Özçelikkale, C. Häger, M. F. Keskin, and H. Wymeersch, “Semantic communication for rate-limited closed-loop distributed communication-sensing-control systems,” arXiv:2512.19177, Dec. 2025.
- [79] X. Fang *et al.*, “Sensing-communication-computing-control closed-loop optimization for 6G digital twin-empowered robotic systems,” *IEEE J. Sel. Areas Commun.*, vol. 43, no. 10, pp. 3330–3346, Oct. 2025.
- [80] G. Pang, W. Liu, D. Niyato, B. Vucetic, and Y. Li, “Communication-control codesign for large-scale wireless networked control systems,” *IEEE J. Sel. Areas Commun.*, vol. 43, no. 10, pp. 3295–3312, Oct. 2025.
- [81] Z. Han, X. Li, Z. Zhou, K. Huang, Y. Gong, and Q. Zhang, “Wireless communication and control co-design for system identification,” *IEEE Trans. Wireless Commun.*, vol. 23, no. 5, pp. 4114–4126, May 2024.
- [82] Z. Han, X. Li, G. Zhu, K. Huang, Y. Gong, and Q. Zhang, “An energy-efficient wireless communication and control co-design for WNCs,” *IEEE Trans. Wireless Commun.*, vol. 25, pp. 1798–1812, 2026.
- [83] E. Li, H. Du, and K. Huang, “NetMCP: Network-aware model context protocol platform for LLM capability extension,” arXiv:2510.13467, Oct. 2025.
- [84] Qwen Team, “Qwen3 technical report,” arXiv:2505.09388, May 2025.
- [85] C. Wu *et al.*, “Janus: Decoupling visual encoding for unified multimodal understanding and generation,” arXiv:2410.13848, Oct. 2024.
- [86] X. Chen *et al.*, “Janus-Pro: Unified multimodal understanding and generation with data and model scaling,” arXiv:2501.17811, Jan. 2025.
- [87] M. Wei *et al.*, “Ground slow, move fast: A dual-system foundation model for generalizable vision-and-language navigation,” in *Proc. ICLR*, 2026.
- [88] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd ed. New York, NY, USA: Springer, 2009, ch. 4, pp. 101–137.
- [89] W. Wang, M. Zhu, J. Wang, X. Zeng, and Z. Yang, “End-to-end encrypted traffic classification with one-dimensional convolution neural networks,” in *Proc. IEEE ISI*, 2017, pp. 43–48, doi: 10.1109/ISI.2017.8004872.
- [90] A. Graves and J. Schmidhuber, “Framewise phoneme classification with bidirectional LSTM and other neural network architectures,” *Neural Networks*, vol. 18, nos. 5–6, pp. 602–610, 2005.